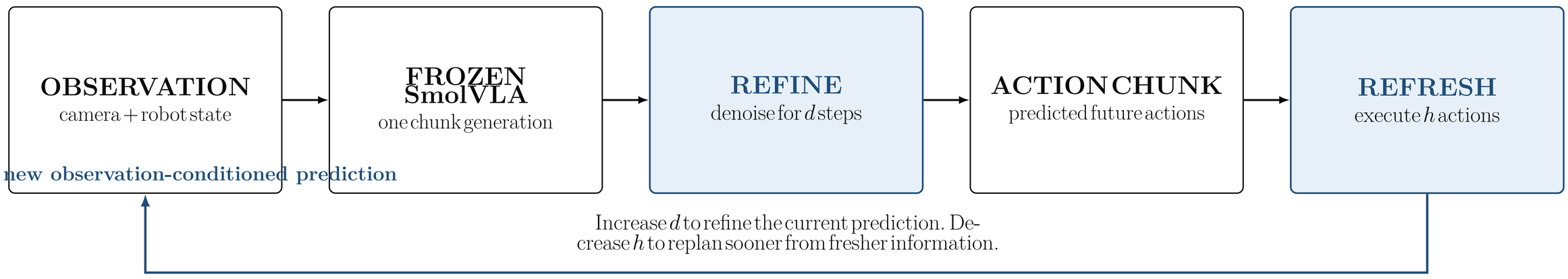
REFINEMENT VERSUS REFRESH

Compute Allocation for Closed-Loop Robot Policies

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A robot can spend compute improving its current action prediction—or use a newer observation to plan again. Which is worth the cost?

CLOSED-LOOP COMPUTE ALLOCATION



01 Study design

A paired discovery study evaluates 12 configurations on four LIBERO Spatial tasks (480 attempts). Seven frozen configurations are then tested on eight distinct Goal and Object tasks excluded from selection (1,120 attempts).

Discovery	12 settings; 4 tasks; 480 attempts
Confirmation	7 frozen settings; 8 held-out tasks; 1,120 attempts

82.1%

of native generation arithmetic is independent of the denoising-step count.
Reducing refinement alone leaves most of the cost of a fresh prediction intact.

02 Compute boundary

For full execution prefixes, the nominal neural arithmetic rate is

$$b(h, d) = \frac{F_{\text{gen}}(d)}{h}.$$

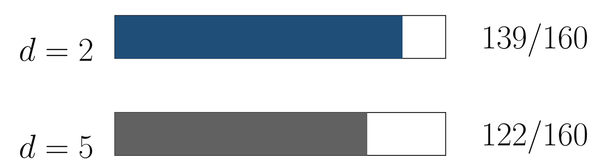
This counts reference-calibrated dense neural FLOPs. It does not measure physical-robot latency, asynchronous overlap, or energy.

Scope: one frozen checkpoint in synchronous simulation. The study does not establish cross-policy generalization or a universal scheduler.

03 Primary result

At a fixed two-action prefix ($h = 2$), compare two versus five denoising steps.

CLOSED-LOOP SUCCESS



+10.6 pp

paired success difference, $d = 2$ minus $d = 5$
Descriptive task-conditional 95% interval: +4.4 to +16.9 percentage points.

−5.9%

reference-calibrated neural arithmetic per control step when reducing d from 5 to 2.

Result: for this checkpoint and task family, additional refinement does not reliably improve closed-loop success.

04 Refresh less often?

Hold denoising fixed at $d = 2$. Compare executing two versus five actions before replanning.

Setting	Success	TFLOPs / attempt
$h = 2,\ d = 2$	139/160	41.1
$h = 5,\ d = 2$	138/160	16.1

−60.7%

arithmetic per attempted episode for $h = 5$ versus $h = 2$, at $d = 2$.

Caution: similar observed success is not evidence of equivalent quality. Non-inferiority or equivalence was not established.

05 Interpretation

- More denoising is not uniformly beneficial.
- Refresh frequency and refinement must be evaluated jointly.
- A simple cost-aware rule matched the fitted model at three compute budgets; fitting showed no demonstrated advantage.

06 Reproducibility

Paired execution and internal outcome checks support the recorded results. Independent model-free replay remains a subsequent validation step.

Paper and code

github.com/amtellezfernandez/

refinement-refresh-robot-policies

TAKEAWAY: Allocate inference compute using closed-loop outcomes, not denoising count alone.